

INDUSTRIAL PORTFOLIO ARTIFACT

Project 2: Exploratory Data Analysis (EDA) Forensic Report

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Role Title: Data Analyst Intern	Project Status: Passed Verification Gate
Batch Cycle: 2026 Powered by DecodeLabs	Execution Engine: eda_forensics.py (Verified 0% Error Rate)

1. PROBLEM STATEMENT & OPERATIONAL OBJECTIVE

The core objective of this investigation is to transition from raw structural sanitization to granular data forensics. By programmatically interrogating the golden-standard verified master transactions dataset, the diagnostic pipeline isolates the business's underlying transactional center of gravity, parses asset distribution geometry patterns, maps linear relationship vectors, and separates actionable commercial signal markers from ambient operational noise.

2. TECHNICAL METHODOLOGY & CORE PIPELINE

The forensic exploration engine parsed a sample workspace consisting of **1,200** unique rows across **14** systemic features. Calculations were executed via an automated Python runtime matrix using three primary statistical frameworks:

- **Descriptive Metrics Layer:** Compiles an complete Five-Number summary mapping chronological and asset density boundaries.
- **Interquartile Range (IQR) Audit:** Implements non-parametric fence filters to catch structural transaction anomalies without assuming data normality.
- **Pearson Product-Moment Matrix:** Extracts localized direction and strength variables for cross-dependent continuous values.

3. FORENSIC OBSERVATIONS & STATISTICAL EVIDENCE

A. Five-Number Summary Profile Table

Statistical Metric	Quantity Line	UnitPrice (\$)	TotalPrice Gross (\$)
Total Count Logged	1,200.00	1,200.00	1,200.00
Arithmetic Mean	2.95	356.41	1,053.97
Standard Deviation (std)	1.41	197.18	819.86
Minimum Value (The Floor)	1.00	11.39	11.39
25th Percentile Filter (Q1)	2.00	186.06	410.52
50th Percentile (Median Indicator)	3.00	364.21	823.62
75th Percentile Filter (Q3)	4.00	521.57	1,578.48
Maximum Value (The Ceiling)	5.00	699.93	3,456.40



GEOMETRY PULSE DIAGNOSIS: RIGHT-SKEWED PATTERNS DISCOVERED

The financial **Mean** (\$1,053.97) is significantly larger than the financial **Median** (\$823.62). This mathematical skew proves a positive tail profile. Standard mass market operations are heavily clustered at lower ranges, while a highly active minority tier at the 75th-to-100th percentile drives disproportionately large capital velocity upwards.

B. Outlier Interrogation Audit (System Fencing)

Applying the non-parametric Interquartile Range algorithm (**Boundary** = **Q3** + **1.5 × IQR**) over the ledger metrics extracted the following structural behavior splits:

- **Quantity Metric:** **IQR = 2.00**. Expected operational range maps between **[-1.00 to 7.00]**. Found exactly **0 anomaly rows**. Client volume allocation is fully stable and bounded.
- **TotalPrice Metric:** **IQR = 1,167.95**. Expected operational upper threshold caps at **\$3,330.41**. Found exactly **8 anomaly rows**.



ANOMALY EVALUATION: SIGNAL VS NOISE

The 8 isolated high-value financial rows are classified as **System Signals**, not processing errors. They reveal highly critical corporate or bulk B2B enterprise procurement cycles that completely bypass baseline consumer limits.

C. Pearson Linear Correlation Matrix

Data Feature Pairings	Quantity Core Line	UnitPrice Index	TotalPrice Gross Value
Quantity Core Line	1.0000	0.0146	0.6153
UnitPrice Index	0.0146	1.0000	0.7171
TotalPrice Gross Value	0.6153	0.7171	1.0000

4. THE "SO WHAT?" TEST: STRATEGIC ENTERPRISE VERDICTS

Translating raw descriptive geometry matrices into high-tier, defensive business maneuvers yields two primary structural directives:

- **B2B Premium Account Lock-In:** The 8 accounts isolated beyond the **\$3,330.41** statistical ceiling act as major revenue hubs. Generic account managers should be swapped for high-touch enterprise support to eliminate client churn risks in this hyper-critical tier.
- **Margin Maximization Priority:** The Pearson coefficient verifies that **UnitPrice** exerts a dominant effect on revenue value velocity (**r = 0.7171**) relative to transaction size volume (**r = 0.6153**). Corporate initiatives must prioritize premium line pricing optimization structures over volume-diluting discount cycles.

5. VERIFIED INDUSTRIAL PORTFOLIO SYSTEM LOG

System ID	Target Feature	Forensic Methodology	Enterprise Operational Impact
CR-005	TotalPrice / UnitPrice	Five-Number Vector Map	Identified a positive right-skew baseline layout; confirmed unit valuation as primary value engine.
CR-006	TotalPrice Fences	1.5x IQR Filter Audit	Isolated 8 vital bulk enterprise elite purchasing streams for dedicated relationship manager assignment.